

worked problems
wed 9 + TH 9

2.4. Applications of Linear First-Order Differential Equations

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In Exercises 14–20, solve each of the given nonhomogeneous linear first-order DEs.

14. $y' + y = 2$

15. $y' + 2y = 2t$

16. $y' + ty = 10t$

17. $y' + \frac{2}{t}y = e^t$

18. $y' = -(y - 1) \cot t$

19. $(1 + t^2)y' + 2ty = 2t$

20. $y' = 0.03 - \frac{2}{100 - t}y$

In Exercises 21–27, solve each of the given initial-value problems.

21. $y' + y = 2, \quad y(0) = 3$

22. $y' + 2y = 2t, \quad y(1) = 0$

23. $y' + ty = 10t, \quad y(0) = 5$

24. $y' + \frac{2}{t}y = e^t, \quad y(1) = 4, \quad t > 0$

✓ 25. $y' = -(y - 1) \cot t, \quad y(\pi/2) = 1, \quad 0 < t < \pi$

sep

26. $(1 + t^2)y' + 2ty = 2t, \quad y(0) = 1$

27. $y' = 0.03 - \frac{2}{100 - t}y, \quad y(0) = 1$

2.4 Applications of Linear First-Order Differential Equations

A large number of important physical situations can be modeled by linear first-order differential equations. In this section we introduce several such applications through examples and explore further scenarios in the exercises.

1. $y' = 10y$

2. $y' = 10y + 10$

3. $y' = 10y^2$

4. $y' = 10y^2 - 10$

5. $t^2y' + y^2 = 1$

6. $e^{3t+y} \frac{dy}{dt} = 1$

7. $t dy - (y - 1) dt = 0$

8. $\frac{dy}{dt} = \frac{5ty - t}{4 + t^2}$

✓ 9. $y - t \frac{dy}{dt} = 6 - 3t^2 \frac{dy}{dt}$

✓ 10. $\frac{dy}{dt} = \frac{-2ty}{t^2 + 1}$

11. $(2 + t^2)y' + 2ty = 0$

12. $3y^2y' + t^2 = 0$

13. $(y + t)y' + y = t$

14. $y' \sin 2t + 2y \cos 2t = 0$

Solve each of the DEs in Exercises 15–28.

15. $y' = 10y$

16. $y' = 10y + 10$

17. $y' = 10y^2$

18. $y' = 10y^2 - 10$

19. $t^2y' + y^2 = 1$

✓ 20. $e^{3t+y} \frac{dy}{dt} = 1$

21. $t dy - (y - 1) dt = 0$

22. $\frac{dy}{dt} = \frac{5ty - t}{4 + t^2}$

✓ 23. $y - t \frac{dy}{dt} = 6 - 3t^2 \frac{dy}{dt}$

24. $\frac{dy}{dt} = \frac{-2ty}{t^2 + 1}$

25. $(2 + t^2)y' + 2ty = 0$

26. $3y^2y' + t^2 = 0$

27. $(y + t)y' + y = t$

28. $y' \sin 2t + 2y \cos 2t = 0$

Solve each of the IVPs stated in Exercises 29–42.

29. $y' = 10y, \quad y(0) = 3$

✓ 30. $y' = 10y + 10, \quad y(0) = 2$

31. $y' = 10y^2, \quad y(1) = 4$

32. $y' = 10y^2 - 10, \quad y(1) = -1$

33. $t^2y' + y^2 = 1, \quad y(2) = 0$

34. $e^{3t+y} \frac{dy}{dt} = 1, \quad y(0) = 0$

35. $t dy - (y - 1)dt = 0, \quad y(1) = 3$

36. $\frac{dy}{dt} = \frac{5ty - t}{4 + t^2}, \quad y(1) = 1$

37. $y - t \frac{dy}{dt} = 6 - 3t^2 \frac{dy}{dt}, \quad y(1) = 5$

38. $\frac{dy}{dt} = \frac{-2ty}{t^2 + 1}, \quad y(0) = 4$

39. $(2 + t^2)y' + 2ty = 0, \quad y(1) = 1$

40. $3y^2y' + t^2 = 0, \quad y(0) = 1$

41. $(y + t)y' + y = t, \quad y(0) = 1$

wed

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$$\frac{dy}{dt} = - (y-1) \frac{\cos(t)}{\sin(t)}$$

sep

$$\int \frac{dy}{y-1} = \int \frac{-\cos(t)}{\sin(t)} dt + C$$

$$\ln|y-1| = - \int \frac{1}{u} du + C \quad \begin{matrix} u = \sin(t) \\ du = \cos(t) \end{matrix}$$

$$\ln|y-1| = - \ln|\sin(t)| + C$$

on



$$\frac{dy}{dt} + y \frac{\cos(t)}{\sin(t)} = \frac{\cos(t)}{\sin(t)}$$

$$v(t) = e^{\ln(\sin(t))} = \sin(t)$$

$$\sin(t) y' + \cos(t) y = \cos(t)$$

$$(y \sin(t))' = \cos(t)$$

$$y \sin(t) = \sin(t) + C$$

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$$e^{3t} \cdot e^y \frac{dy}{dt} = 1 \quad \text{sep}$$

$$\int e^y dy = \int e^{-3t} dt + C$$

$$e^y = \frac{e^{-3t}}{-3} + C$$

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~~Answer~~

$$y + (3t^2 - t) \frac{dy}{dt} = 6$$

lin

$$\frac{dy}{dt} + \frac{1}{3t^2 - t} y = \frac{6}{3t^2 - t}$$

$$v(t) = e^{\int \frac{1}{3t^2 - t} dt} \quad \text{☹️}$$

sep

$$(3t^2 - t) \frac{dy}{dt} = 6 - y$$

$$\int \frac{dy}{6-y} = \int \frac{dt}{3t^2 - t} + C$$

$$-\ln(6-y) = \int \frac{dt}{3t^2 - t} + C$$

Int continued on next page

Int for #23 (easier way)

$$\int \frac{dt}{t(t-3)} = \int \frac{A}{t} dt + \int \frac{B}{t-3} dt$$

where $A/t + B/t-3 = 1/t(t-3)$

~~$$A(t-3) + Bt = 1$$~~

$$A(t-3) + Bt = 1$$

$$(A+B)t - 3A = 0t + 1$$

$$-3A = 1 \quad A = -1/3$$

$$A+B=0 \quad B = -A = 1/3$$

$$\begin{aligned} \int \frac{dt}{t(t-3)} &= -\frac{1}{3} \ln(t) + \frac{1}{3} \ln(t-3) \\ &= \frac{1}{3} (\ln(t-3) - \ln(t)) \end{aligned}$$

$$\int \frac{1}{t(t-3)} dt = \frac{1}{3} \ln(t-3/t) = \ln \left(\frac{t-3}{t} \right)^{1/3}$$

p137

TH

9

~~TH~~

$$y - t y' = 6 - 3t^2 y'$$

check exact $dy - t dx = 6 dt - 3t^2 dy$

$$(y-6)dt + (3t^2-t)dy = 0$$

not exact $\frac{\partial}{\partial y}(y-6) = 1 \neq \frac{\partial}{\partial t}(3t^2-t)$

TH

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$$y' - 10y = 10 \quad y(0) = 2 \quad \text{lin}$$

$$e^{-10t} y' - 10e^{-10t} y = 10e^{-10t} \quad v = e^{-10t}$$

$$y e^{-10t} = \int 10 e^{-10t} dt + C$$

$$u = -e^{-10t} + C$$

$$2e^0 = -e^0 + C \Rightarrow \boxed{C=3}$$

$$\boxed{y e^{-10t} = -e^{-10t} + 3}$$

p137

Th

10

$$dy/dt = -2ty / (t^2 + 1)$$

check exact

$$2ty dt + (t^2 + 1) dy = 0$$

$$\frac{\partial}{\partial y}(2ty) = 2t = \frac{\partial}{\partial t}(t^2 + 1) \quad \text{Exact.}$$

$$\frac{\partial \phi}{\partial t} = 2ty \Rightarrow \phi = t^2 y + h(y)$$

$$\Rightarrow \frac{\partial \phi}{\partial y} = t^2 + h'(y) = t^2 + 1$$

$$\boxed{t^2 y + y = C}$$

$$h'(y) = 1 \quad h(y) = y$$

■ Solve $y' = y + ty^2$

$$\boxed{\begin{aligned} u &= y^{-1} \\ u' &= -y^{-2} y' \end{aligned}}$$

$$\frac{e^t}{y} = C - [te^t - e^t]$$

$$y' - y = ty^2 \Rightarrow y^{-2}y' - y^{-2}y = t$$

$$y^{-2}y' - y^{-1} = t$$

$$-u' - u = t$$

$$e^t(u' + u) = -t$$

$$(e^t u)' = -te^t$$

$$e^t u = \int -te^t dt + C$$

2. Solve $y' + y = y^3$

$$y^{-3}y' + yy^{-3} = 1$$

$$y^{-3}y' + y^{-2} = 1$$

$$-\frac{1}{2}u' + u = 1$$

$$u' - 2u = -2 \quad \text{lin} \quad v = e^{-2t}$$

$$e^{-2t}u' - 2ue^{-2t} = -2e^{-2t}$$

$$(e^{-2t}u)' = -2e^{-2t}$$

$$e^{-2t}u = -\int e^{-2t} dt + C$$

$$e^{-2t}u = \frac{1}{2}e^{-2t} + C$$

$$\boxed{e^{-2t}y^{-2} = \frac{1}{2}e^{-2t} + C}$$

$$\boxed{\begin{aligned} u &= t \quad dv = e^t dt \\ \int u dv &= uv - \int v du \\ du &= dt \quad v = e^t \end{aligned}}$$

$$\boxed{u = y^{-2}, \quad u' = -2y^{-3}y'}$$